APPLY YOUR KNOWLEDGE

In what follows, the value of the gravitational acceleration is: $g = 10\ m/s^2$

3 Write the expression for the variation of the total energy of a system in the two following cases:

a) U = constant, E_m varies

b) U varies, E_m = constant.

4 Using the principle of the conservation of mechanical energy, show that the acceleration $\vec{a}$ of a freely falling body has the value g.

5 A small ball of mass m = 200 g is placed on a platform fixed to a wall close to a well (Figure 1.14). Determine the gravitational potential energy of the system (ball, Earth), taking its reference level successively as the horizontal plane containing:

a) The bottom of the well.

b) The ground level.

c) The platform.

d) The top of the wall.

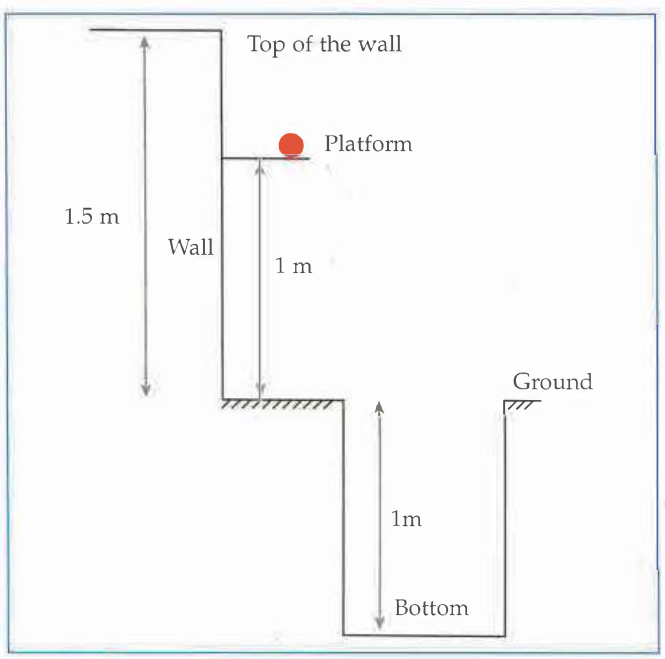

Figure 1.14

6 A particle B of mass m = 10 g is shot vertically upward from a point O with velocity v_0= 10 m/s. The horizontal plane containing O is the reference level of the gravitational potential energy. Neglect all air effects.

a) Calculate the mechanical energy of the system (B, Earth) at O.

b) Calculate the mechanical energy of the system (B, Earth) at the highest point A. Interpret the result.

7 A puck is released on an air table inclined by α = 0.1 rd with respect to the horizontal. By applying the principle of conservation of the mechanical energy of the system (puck, Earth), calculate the velocity of the puck after a 40 cm journey along a line of greatest slope.

8 A particle (C) of mass m = 20 g can move without friction along a slide placed in a vertical plane. The slide is composed of two rails: one OA is horizontal, and the other AB is inclined with respect to the horizontal (Figure 1.15).

C is launched from O with an initial velocity $\vec{v}_0$ whose magnitude is v_0 = 5 m/s. Taking the horizontal plane containing O as a reference level for the gravitational potential energy:

a) Calculate the mechanical energy of the system (C, Earth) at point O, and at the highest point B reached by (C).

b) Calculate the altitude of this point B.

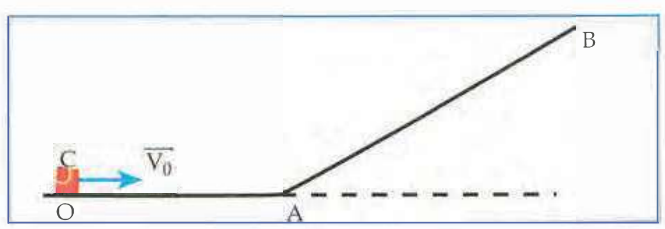

Figure 1.15

9 A homogeneous rod, of mass M = 20 kg, of length L = 3 m, and of negligible cross-section, rests initially on the horizontal ground chosen as a reference level for the gravitational potential energy. The rod is rotated about one end until it takes a vertical position. Calculate the variation of the mechanical energy of the system (rod, Earth).

10 A particle C, of mass 2 kg, is released without initial velocity from the top A of an inclined plane making an angle of 30° with the horizontal (Figure 1.16). The length of the line of greatest slope passing through A is AB = 2.5 m. C reaches point B with avelocity of 3 m/s. Take as the reference level of gravitational potential energy the horizontal plane passing through B. Calculate the mechanical energy of the system (C, Earth) at point A, and at point B. Deduce its variation. What do you conclude?

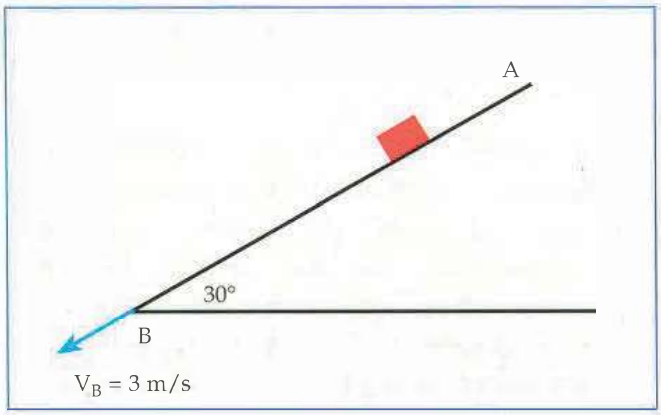

Figure 1.16

11 A particle C can move along a slide placed in a vertical plane (Figure 1.17). Points A and B are at the same horizontal level. Determine, with respect to B, the point where the velocity of C is zero for the first time in the three following cases:

a) C is released from A without initial velocity (Neglect friction).
b) C is launched downward from A along the slide (Neglect friction).
c) C is released from A without initial velocity but is subjected to a frictional force.

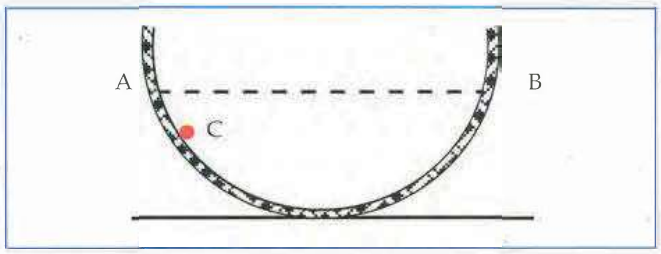

Figure 1.17

12 A lake of water is at a height of 80 m above a turbine-alternator system that furnishes an electric power of 10^5 kW.

a) In what form is the energy of the lake stored?
b) Knowing that the efficiency of the water-turbine-alternator system is 40%, calculate the mass flow rate of water.

13 A hoop (C), of mass m = 400 g and of radius R = 30 cm, can rotate about a horizontal axis (Δ) passing through a point A of its circumference (Figure 1.18). Take the horizontal plane passing through A as a reference level for the gravitational potential energy.

a) Calculate, at equilibrium, the gravitational potential energy of the system ((C), Earth).
b) Calculate the gravitational potential energy of the system ((C, Earth) when (C) is shifted by an angle of 60° about (Δ). Deduce the variation of the gravitational potential energy of the system.
c) The hoop is left without initial velocity. Calculate, by applying the principle of conservation of mechanical energy, the

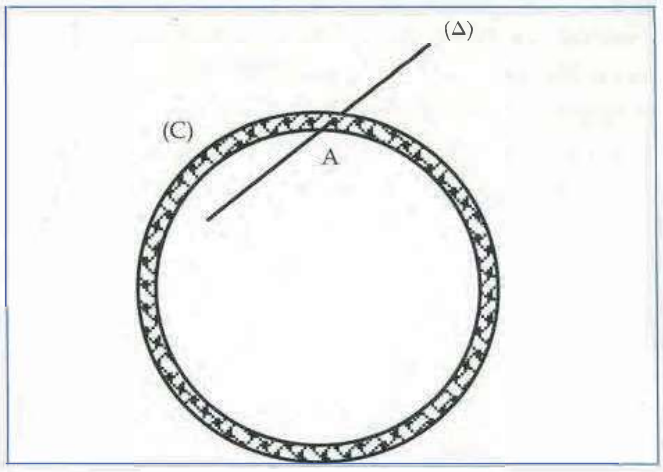

Figure 1.18

angular velocity of the hoop as it passes through the equilibrium position.

The moment of inertia of (C) with respect to (Δ) is $I = 2mR^2$.

14 Two particles A and B, of respective masses m_A and m_B, are connected together by means of an inextensible string of negligible mass. The string passes over the groove of a pulley of negligible mass as shown in the figure 1.19.

At instant $t_0 = 0$, the system S (pulley, mass, string) is left without initial velocity, with particle B being at a distance h from the axis of the pulley. The reference level of gravitational potential energy is that of a horizontal plane containing the axis of the pulley and the particle A. (Neglect friction).

a) Calculate, at t = 0, the mechanical energy of the system (S, Earth) in terms of m_B, g, and h.

b) Calculate the mechanical energy of the system (S, Earth) after covering a distance x.

c) Calculate, by applying the principle of the conservation of mechanical energy, the velocity v of A or B in terms of x, m_A, m_B and g. Deduce the acceleration of the motion.

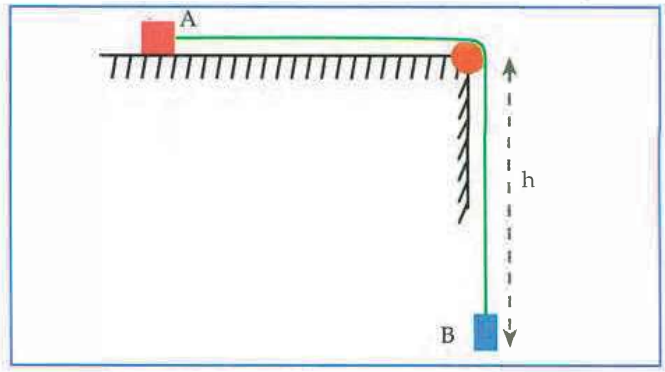

Figure 1.19

PROBLEMS

15 A parachutist (S), of total mass 100 kg, jumps from an airplane flying at an altitude of 1000 m with a speed of 270 km/h with respect to the ground. (S) reaches the ground with a speed of 18 km/h.

a) Is the mechanical energy of the system ((S), Earth) conserved during the jump? Interpret.

b) Calculate the thermal energy produced.

16 A pendulum is formed of a particle, of mass m = 100 g, suspended from the end of an inextensible string of negligible mass and of length L = 100 cm. The pendulum is shifted by an angle of 60° from its equilibrium position about a horizontal axis passing through the other end of the string, and released without initial velocity. The string is still stretched. Take the reference level of the gravitational potential energy as that of a horizontal plane containing the equilibrium position of the particle. Neglect friction.

a) Calculate the mechanical energy of the system S (pendulum, Earth, support).

b) Find, in terms of θ ($\theta < 60°$), the expressions for the kinetic and the potential energies of the system S.

c) The graphs 1, 2, and 3 of figure 1.20 represent energies of S. What type of energy does each graph represent?

Scale:

- 1 cm on the x - axis (θ) corresponds to 10°.

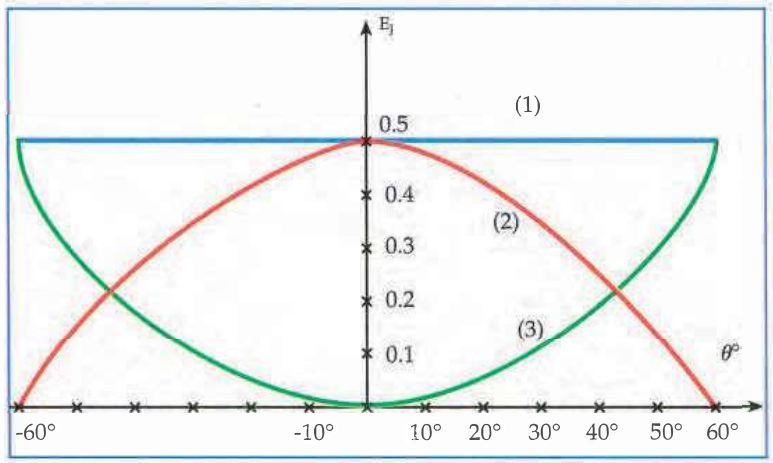

Figure 1.20

- 1 cm on the y axis (Energies) corresponds to 0.1 J.

d) Find graphically the positions of the pendulum for which the kinetic energy is equal to the gravitational potential energy.

17 To launch a solid of mass m on an inclined slide, we use the apparatus shown in figure 1.21. The natural length of the spring is L_0. Before launching, the spring is compressed to a length L. After launching, the center of mass of the solid reaches the point A of altitude Z with a velocity v. Neglect friction.

a) Determine the relation among L_0, L, m,v, Z, g, and the constant k of the spring.

b) Calculate k if the maximum altitude reached is 20 cm. Take m = 600 g, L_0 = 25 cm, and L = 20 cm.

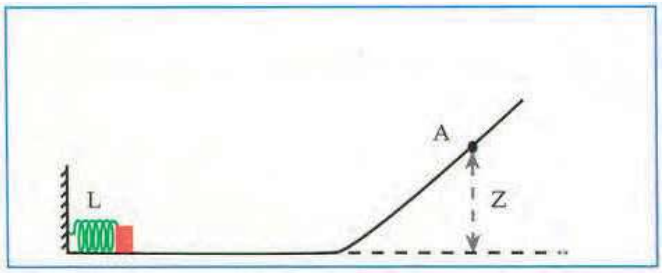

Figure 1.21

18 A rope, of mass M uniformly distributed along its length L, can slide without friction on a horizontal table. At instant $t_0 = 0$, the hanging part of the rope has a length b, and the rope is held at rest (Figure 1.22).

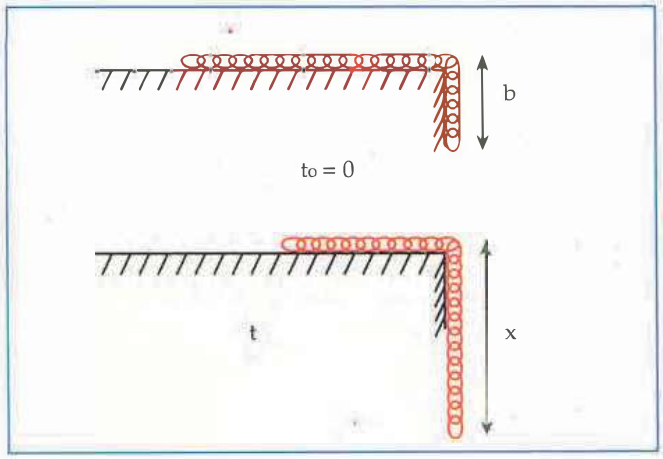

Figure 1.22

When it is left, the rope begins to slide. At instant t, the part of the rope that hangs has a length x and a velocity v. Take the level of the horizontal table as reference level for the gravitational potential energy.

a) Calculate, at instant $t_0 = 0$, the mechanical energy of the system (rope, Earth, table) in terms of M, L, b, and g.

b) Calculate, at instant t, the mechanical energy of the system (rope, Earth, table) in terms of M, L, x, v, and g.

c) Find a relation between x and v.

d) Find, at instant t, the acceleration of the motion of the rope in terms of x, L, and g. Deduce the value of this acceleration for x = L.

19 A particle (C) of mass m = 200 g, can slide without friction on an inclined plane making an angle $\alpha = 30°$ with respect to the horizontal. (C) is launched, at instant $t_0 = 0$, along a line Ox of greatest slope with a velocity of $V_0 = 5$ m/s (Figure 1.23). Thus, (C) moves along Ox. At any instant t, the position of (C) is given by its abscissa x = OA. The horizontal plane passing through O is the reference level for the gravitational potential energy.

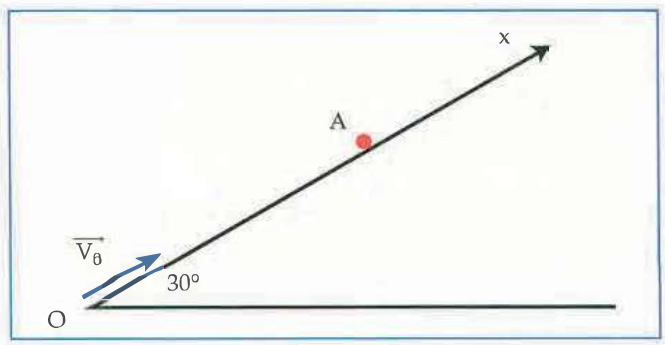

Figure 1.23

a) Calculate the mechanical energy of the system {(C), Earth}.

b) Express, in terms of x, the gravitational potential energy of the system {(C), Earth}.

c) Sketch the graph of the mechanical energy and that of the gravitational potential energy. Use the following scales:

- 1 cm corresponds to 0.5 m of distance on the abscissa axis.
- 1 cm corresponds to 0.5 J of energy on the ordinate axis.

d) Determine, graphically, the position of (C) for which the velocity is V = 2 m/s.

e) Determine the maximum altitude reached by (C).